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Semantic-Guided GAN Framework for Generating Adverse Weather Conditions from Clear-Weather Images

**Semantisch gesteuertes GAN-Framework zur Generierung ungünstiger
Wetterbedingungen aus Bildern bei klarem Wetter**

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This work explores the task of semantic image synthesis under adverse weather conditions using an extended DP-GAN framework guided by semantic segmentation maps and clear weather reference images. We propose a novel fusion strategy for integrating clear-weather features into the generative process using dual SPADE normalization with a learnable fusion mechanism, embedded within a three-pyramid generator architecture. To evaluate the effectiveness of this approach, the generated images are employed as data augmentation to train a semantic segmentation model, enabling an assessment of their semantic consistency and downstream utility. Among all tested baseline, our proposed methodology yielded the best performance, achieving the lowest FID score of 98.01 and demonstrating superior perceptual quality and semantic alignment.

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1 Introduction

In the domain of autonomous systems, the ability to accurately perceive and react to diverse environmental conditions is crucial for safety and efficiency. Object detection is central to these systems as it enables the reliable identification of obstacles and entities essential for navigation. However, a significant gap is that object detection models are often trained on datasets that lack images with adverse weather conditions, such as fog, rain, and snow. As a result, these models struggle to perform accurately in challenging weather conditions. This research addresses this gap by proposing a framework to synthesize images with adverse weather conditions, specifically fog conditions, using generative adversarial networks guided by semantic segmentation maps, thus enhancing dataset diversity and supporting the creation of more robust, weather-resilient object detection models.

1.1 Motivation

Existing approaches for adverse weather generation often fail to maintain spatial and semantic consistency across translated images. For example, when translating a clear day scene featuring a busy street into a foggy day scene, the translated image must depict the same street layout, vehicles, and buildings i.e., cars should remain clearly identifiable as cars, and their spatial relations (e.g., a vehicle behind a traffic light) should remain unchanged, preserving both the object identity and the scene's depth. Moreover, the need for high realism in synthetic images, with accurate representations of complex weather phenomena like fog density and rain reflections, remains unmet. For example, fog tends to be denser farther from the viewer and lighter closer; a wet road should reflect street lights or neon signs realistically, including the diffusion of light due to water. These challenges highlight the necessity for a semantic image synthesis model to control the environmental variables more precisely and adhere to real-world visual principles.

1.2 Problem Statement

As shown in Figure 1.1, the goal is to develop a model capable of simulating realistic weather effects without direct exposure to weather images during inference. The model can generalize weather transformations and create diverse, data-efficient augmentations by conditioning only on clear-weather reference images and semantic segmentation maps. This approach can then reduce the need for expensive data collection and annotation in adverse conditions while improving the robustness and generalization of downstream object detection models in real-world environments.

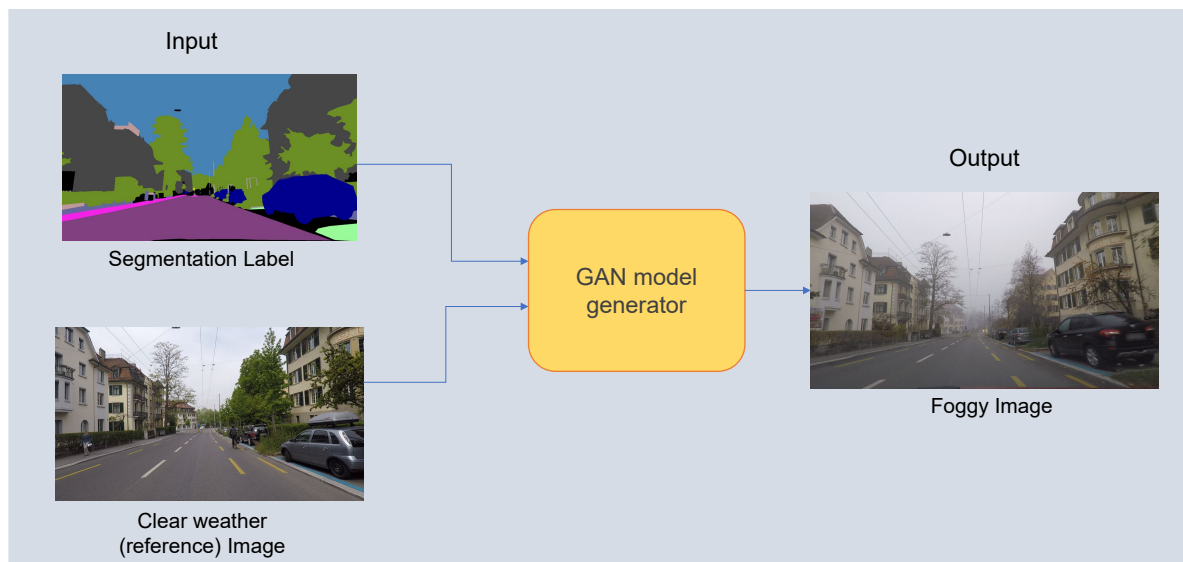


Figure 1.1: Our proposed methodology structure. Our proposed model takes a clear image (lower left) along with its segmentation label (upper left) as input and generates a realistic image with the weather condition as output.

2 Literature Review

This section provides an overview of key concepts and models that form the foundation for the methods explored in this research. It introduces the basic principles of Generative Adversarial Networks (GANs), along with notable variants and models. These models and techniques are referenced throughout the thesis and are essential for understanding the approach to generating adverse weather conditions from clear weather images.

2.1 Generative Adversarial Networks

One of the most prominent and widely adopted technologies for generative models is *Generative Adversarial Networks* (GANs) introduced by [1]. At their core, GANs consist of two neural networks: the generator and the discriminator. Together, they engage in a dynamic, adversarial game. This interplay allows GANs to learn how to generate new data samples that are statistically similar to a given training dataset.

As shown in Figure 2.1, the generator network tries to create synthetic data (e.g., images, audio, text) that resembles real data. It starts with random noise and gradually learns to produce outputs that could plausibly come from the true data distribution. While the discriminator acts as a critic, distinguishing between real samples from the training data and fake samples produced by the generator.

The two networks are trained simultaneously. The generator improves by learning to "fool" the discriminator, while the discriminator gets better at spotting the difference between real and fake data. Mathematically, this process can be viewed as a minimax game, with the generator minimizing the discriminator's ability to detect fakes and the discriminator maximizing it. The original objective function is expressed in ??.

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (2.1)$$

Here x represents real data samples drawn from the true data distribution $p_{\text{data}}(x)$, z is a random noise vector sampled from a prior distribution $p_z(z)$ (typically uniform or Gaussian), $G(z)$ denotes the generator's output given input noise z , and $D(x)$ represents the discriminator's estimated probability that x is a real data sample rather than a synthetic one.

Training proceeds iteratively: the discriminator is trained to maximize its classification accuracy, while the generator is trained to minimize the discriminator's ability to correctly classify its outputs. Ideally, this adversarial training continues until the generator produces outputs that are indistinguishable from real data, at which point the discriminator cannot perform better than random guessing.

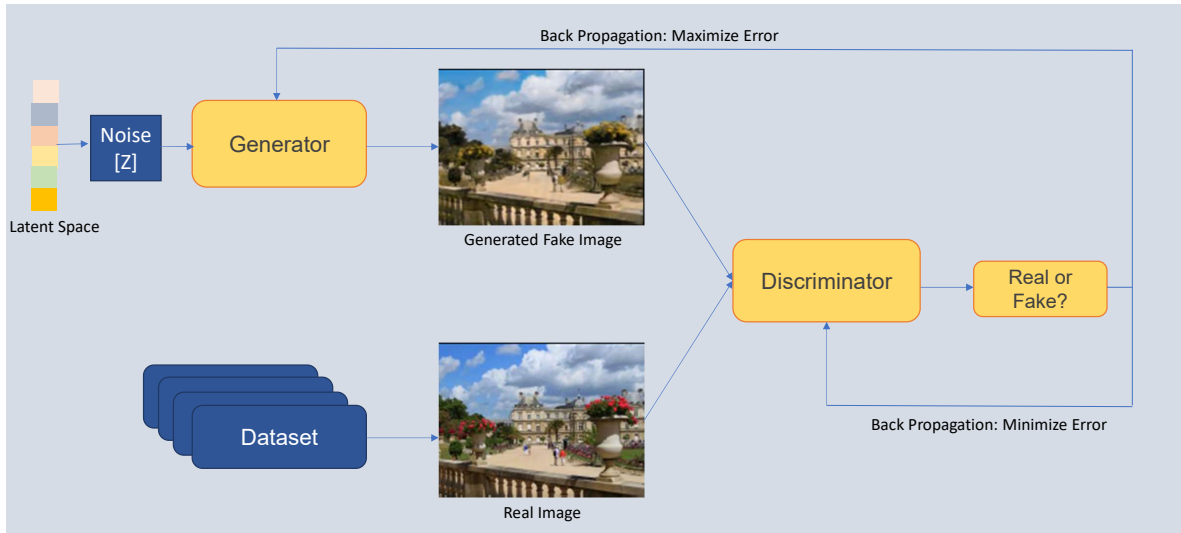


Figure 2.1: The architecture of GAN. The generator uses random noise (upper left) to generate fake images. The discriminator is provided with both real and fake generated images (center) and tries to distinguish the fake ones, while the generator uses the discriminator’s decision to generate fake images as realistically as possible.

One of the key insights of GANs is that they do not explicitly define or estimate the underlying data distribution. Instead, the generator implicitly learns this distribution through the feedback provided by the discriminator.

Conditional GANs (cGANs), an extension of this architecture, condition the generation process on auxiliary information such as class labels, segmentation maps, or input images [2]. As a result, GANs and their variants have become foundational models for generating high-quality, photorealistic images in a wide range of applications, including weather simulation [3, 4], domain adaptation [5, 6], and semantic-to-realistic image synthesis [7].

2.2 Image-to-image Translation

Image-to-image translation is a task in computer vision where an input image is transformed into a corresponding output image from a different domain, while preserving the underlying structure, for example, translating a daytime scene into a nighttime version [2].

Early approaches to image-to-image translation that leveraged GAN models include the widely known *pix2pix* framework [8]. This model learns a direct mapping from an input image to a corresponding output image using paired datasets, where each input image is aligned pixel-by-pixel with its target counterpart. While effective in tasks like sketch-to-image or segmentation-to-photo synthesis, this requirement for precisely aligned data limits their applicability, especially in domains where collecting such data is costly, time-consuming, or impractical, such as adverse weather conditions or medical imaging.

To overcome the dependency on paired training data, [9] introduced *Cycle-Consistent Adversarial Networks* (CycleGAN), a framework that enables **unpaired image-to-image translation**. Unlike conditional GANs (cGANs), which rely on corresponding input-output image

pairs, CycleGAN learns to translate between two domains $\mathcal{X}$ and $\mathcal{Y}$ using **unpaired** samples. This makes it particularly useful in scenarios where paired datasets are unavailable or difficult to collect.

CycleGAN uses two **generators**, $G : \mathcal{X} \rightarrow \mathcal{Y}$ and $F : \mathcal{Y} \rightarrow \mathcal{X}$, along with two **discriminators**, D_Y and D_X , each responsible for distinguishing real images from synthetic images in their respective domains.

The training is guided by two key loss components:

- **Adversarial Loss 2.2** This loss encourages each generator to produce outputs indistinguishable from real images in the target domain. For the mapping $G : \mathcal{X} \rightarrow \mathcal{Y}$, the adversarial loss is defined as:

$$\mathcal{L}_{\text{GAN}}(G, D_Y, \mathcal{X}, \mathcal{Y}) = \mathbb{E}_{y \sim p_{\text{data}}(y)}[\log D_Y(y)] + \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log(1 - D_Y(G(x)))] \quad (2.2)$$

A similar adversarial loss is applied for the reverse mapping $F : \mathcal{Y} \rightarrow \mathcal{X}$:

$$\mathcal{L}_{\text{GAN}}(F, D_X, \mathcal{Y}, \mathcal{X}) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D_X(x)] + \mathbb{E}_{y \sim p_{\text{data}}(y)}[\log(1 - D_X(F(y)))] \quad (2.3)$$

- **Cycle-Consistency Loss 2.4** A novel contribution of CycleGAN is the cycle-consistency loss, which enforces that the mappings G and F are approximately inverses of each other. Specifically, an image translated to the other domain and then back again should return to its original form:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{\text{data}}(y)}[\|G(F(y)) - y\|_1] \quad (2.4)$$

This loss ensures that structural content is preserved during translation, which is critical for many applications.

Total Objective 2.5 The full CycleGAN objective function combines the adversarial and cycle-consistency losses:

$$\mathcal{L}(G, F, D_X, D_Y) = \mathcal{L}_{\text{GAN}}(G, D_Y, \mathcal{X}, \mathcal{Y}) + \mathcal{L}_{\text{GAN}}(F, D_X, \mathcal{Y}, \mathcal{X}) + \lambda \mathcal{L}_{\text{cyc}}(G, F) \quad (2.5)$$

Where λ is a hyperparameter that controls the relative importance of the cycle-consistency loss. By jointly minimizing this objective, CycleGAN can learn bidirectional mappings between domains without requiring paired images.

2.3 Semantic image synthesis

Semantic image synthesis is the task of generating photorealistic images from structured semantic inputs such as segmentation maps or scene layouts [10]. Segmentation maps define the spatial arrangement and class labels of objects within a scene, serving as a high-level blueprint that guides the image generation process. The objective is to synthesize images

that are both visually plausible and semantically faithful to the input. This task is particularly useful in applications such as data augmentation, simulation-to-reality transfer, and controllable image editing. Recent advances in generative adversarial networks (GANs) have significantly improved the performance of semantic image synthesis models. For instance, Pix2PixHD [11] introduced a coarse-to-fine generator and multi-scale discriminator setup to generate high-resolution images from segmentation maps. However, in Pix2PixHD, the segmentation map is typically fed into the network once at the input layer, and its influence diminishes as the image passes through deeper layers. This often leads to loss of semantic alignment and degraded control over the spatial layout of generated images.

To overcome this, *Spatially-Adaptive Normalization* (SPADE) [10] proposed a new normalization layer. The architecture is illustrated in Figure 2.2, where each SPADE ResBlock (ResBlk) injects the semantic layout at every layer of the generator instead of just at the input. Unlike standard normalization methods (like BatchNorm or InstanceNorm), which tend to wash out spatial information, SPADE modulates the activations based on the semantic label map using spatially varying scale and bias parameters.

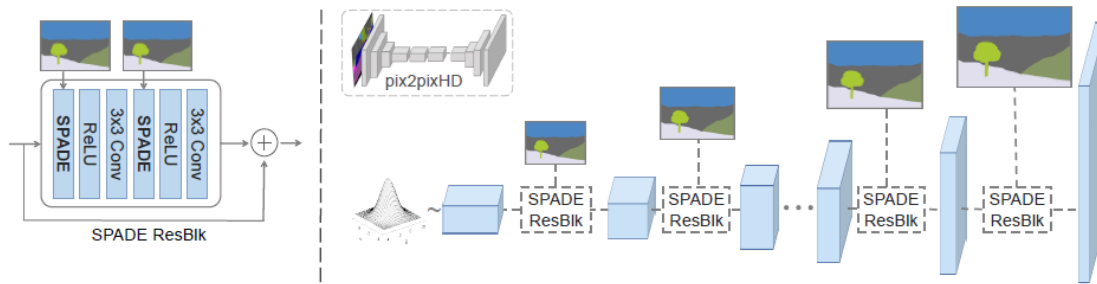


Figure 2.2: In the SPADE generator, each normalization layer uses the segmentation mask to modulate the layer activations. Structure of one residual block with the SPADE (left). The generator contains a series of SPADE residual blocks with upsampling layers (right). This architecture achieves better performance with a smaller number of parameters by removing the downsampling layers of leading image-to-image translation networks such as the Pix2PixHD [8]. Figure adapted from [10].

This architecture allows the generator to preserve fine-grained semantic details and spatial structure throughout the entire network, resulting in images that are not only visually realistic but also semantically accurate.

While SPADE focuses on injecting semantic layout into the generator, it still relies on a traditional discriminator that evaluates images holistically with a binary decision - real/fake. To further enhance semantic fidelity and local detail, newer architectures have introduced more discriminator-centric innovations.

One such advancement is *Only Adversarial Supervision for Semantic Image Synthesis* (OA-SIS) [12], which presented a fully convolutional, spatially adaptive discriminator that performs pixel-level classification, treating semantic image synthesis as a per-pixel classification task rather than a binary decision- real/fake. The discriminator is trained to predict the semantic class of each pixel, allowing it to provide highly localized and class-aware feedback

to the generator. This means that the generator learns not only to produce realistic textures but also to respect the semantic boundaries and characteristics of each object class, whether it's the texture of a road, the transparency of glass, or the haziness of fog. To support this design, OASIS introduces a novel **discriminator loss function** based on a per-pixel cross-entropy loss rather than a binary classification loss. The discriminator loss 2.6 contains a class-weighted cross-entropy loss to address the imbalanced distribution of semantic classes when some object categories appear less frequently or occupy smaller spatial areas and to distinguish between real and generated samples, all pixels in fake images are assigned to an additional class $N + 1$, where N is the number of semantic classes. This allows the discriminator to identify semantic content and recognize synthetic artifacts.

The OASIS discriminator loss is defined as:

$$\mathcal{L}_D = -\mathbb{E}_{(x,t)} \left[\sum_{c=1}^N \alpha_c \sum_{i,j} t_{i,j,c} \log D(x)_{i,j,c} \right] - \mathbb{E}_{(z,t)} \left[\sum_{i,j} \log D(G(z,t))_{i,j,c=N+1} \right] \quad (2.6)$$

where x is a real image with its ground-truth semantic label map t , and $G(z, t)$ is the fake image generated from noise z and label map t . The output of the discriminator $D(\cdot)$ is a per-pixel probability distribution over $N + 1$ classes. The class-balancing weights α_c are inversely proportional to the frequency of class c in the dataset, helping the model to focus more on rare classes. This formulation enables the generator to receive fine-grained, class-aware training signals from the discriminator, improving both realism and semantic accuracy of the synthesized images.

One of the most notable works that utilized both the SPADE normalization and the OASIS discriminator for the task of semantic image synthesis is the *DualPyramid GAN* (DP-GAN) [13]. This model effectively integrates the core principles of these two frameworks to address the challenges of generating high-quality, semantically consistent images.

- **Generator: Dual-Pyramid Structure**

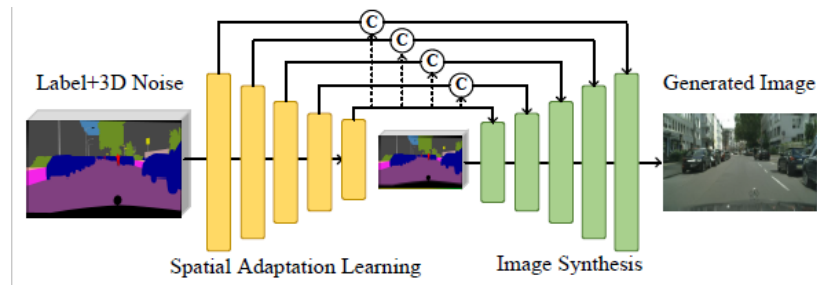


Figure 2.3: The dual pyramid generator architecture. Figure adapted from [13]

As shown in Figure 2.3, the dual pyramid generator follows a two-stage structure composed of spatial adaptation learning and image synthesis, each responsible for capturing different levels of abstraction within the generation process. In the spatial adaptation learning stage (yellow blocks), the model processes the input semantic label map combined with a noise vector to generate a hierarchy of features at multiple spatial scales. In this stage, the generator learns to represent both coarse global structures and

fine local details. The multi-scale features are then propagated through skip connections (denoted by the circular C blocks in Figure 2.3) to the corresponding layers in the image synthesis stage (green blocks in Figure 2.3), which progressively upsamples and fuses them to generate the final image.

The generator uses SPADE layers within both stages to inject semantic information directly into the feature maps at each layer, ensuring strong alignment between the input layout and the generated image.

- **Discriminator: Multi-Scale OASIS**

The discriminator architecture is illustrated in Figure 2.4. It follows a multi-scale, class-aware design inspired by the OASIS framework, providing detailed supervision at different abstraction levels. The model takes both real and generated images, along with their corresponding semantic label maps, and performs real/fake classification at three levels: patch, feature, and pixel.

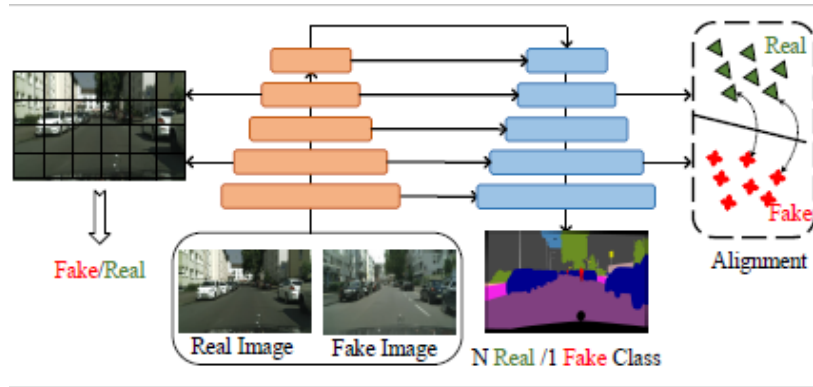


Figure 2.4: The dual pyramid discriminator architecture. Figure adapted from [13].

At the patch level (left side of figure 2.4), the discriminator evaluates local image regions to detect fine-grained inconsistencies. The input is then processed through a pyramid of feature encoders (middle of figure 2.4), which capture both local and global structures. At the pixel level (bottom right of figure 2.4), the discriminator performs class-aware predictions, where real pixels are classified according to their semantic class, and generated pixels are assigned a distinct "fake" class. Together, these multi-scale and class-conditioned objectives allow the discriminator to provide rich feedback, guiding the generator toward producing visually coherent and semantically consistent images.

While image-to-image translation typically relies on visual cues from the input image, semantic image synthesis emphasizes the semantic structure. Our research merges the two concepts by using both a clear-weather image (as a visual reference) and its corresponding segmentation map (as semantic guidance) to synthesize images under foggy weather conditions.

2.4 Weather Image Translation

Several methods have been proposed to address the task of synthesizing adverse weather conditions (such as rain, fog, or snow) from clear weather images. These methods can broadly be divided into two main categories: *analytical methods* and *learning-based methods*, particularly those based on GANs. Analytical methods rely on computer vision and physics-based models to simulate weather effects by explicitly modeling factors such as light scattering or attenuation. While interpretable and often grounded in real-world physical phenomena, these methods typically lack the flexibility and realism achievable with data-driven techniques. On the other hand, GAN-based approaches have emerged as powerful tools for this task, leveraging large datasets to learn weather patterns and generate high-fidelity transformations. These models typically view the task either as a form of *style transfer*, where the appearance of adverse weather is transferred from one domain to another, or as *data augmentation*, where realistic weather effects are directly synthesized to enrich training datasets. Both perspectives will be represented in this section.

2.4.1 Analytical Methods

Early fog simulation methods primarily rely on depth maps and the atmospheric scattering model, where the scene radiance attenuates exponentially with distance. However, these approaches often struggle to replicate the visual complexity and depth-dependent variability observed in real foggy or hazy scenes. To address this, the authors in [14] propose incorporating elevation information along with depth to more accurately model visibility degradation, especially for outdoor environments like urban or mountainous landscapes. *Foggy and Hazy Image Simulator* (FoHIS) simulates fog intensity that varies not just with distance from the camera, but also with terrain height, leading to more visually plausible results. The approach is physics-informed, relying on well-established physical models and handcrafted rules to simulate fog or haze effects, which makes them less adaptive.

[15] compared the following three adverse weather augmentation strategies:

- **1. Rain Simulation:** The authors simulate rain by overlaying streak-like textures on the original image to mimic raindrops falling at an angle. This is done analytically by:
 - Generating synthetic rain streaks using line kernels applied at random positions and angles,
 - Adjusting the intensity and transparency of the rain to simulate varying rainfall densities,
 - Optionally applying motion blur to the rain streaks to match camera exposure effects and the natural motion of rain.

This method creates a visually plausible rain effect without altering the underlying scene geometry or semantics.

- **2. Fog Simulation:**

For fog, the authors use a depth-dependent atmospheric scattering model. The fog simulation results in reduced contrast, desaturation, and a whitening effect, especially at greater scene depths.

- **3. Snow Simulation:**

To simulate snow, the authors added snowflake textures onto the image in multiple layers:

- Foreground layer, flakes are larger and more opaque to mimic snow close to the camera,
- Background layer, flakes are smaller and lighter, creating a depth illusion,

This layered composition creates a volumetric snowfall appearance, mimicking the dynamic and partially occluding nature of real snow.

While these analytical methods provide an interpretable and computationally efficient way to simulate adverse weather conditions, they have several limitations. First, the simulated effects are based on handcrafted models and assumptions that may not fully capture the complexity and variability of real-world weather phenomena. For example, fog simulation assumes a uniform scattering model, which may not reflect localized variations in density or illumination. Similarly, rain and snow effects are often overlaid as 2D textures without accounting for scene geometry, depth occlusions, or dynamic interactions. Another limitation is that physics-based models are typically standalone simulators; they are not optimized jointly with downstream tasks (e.g., segmentation or detection). On the other hand, deep learning methods can be trained end-to-end for task-specific goals or integrated into pipelines that directly optimize perceptual quality. Additionally, the lack of realism in these synthetic effects can limit the generalization of models trained on such data when applied to real-world adverse weather scenarios. Compared to data-driven approaches, these methods offer less adaptability and may underperform in capturing subtle visual cues present in real environmental conditions.

2.4.2 Learning-based Methods

With the advancement of deep generative models, learning-based methods have become the dominant approach for simulating adverse weather conditions in images. In this context, the task of generating adverse weather conditions can be formulated in one of two ways: **(a)** as a *style transfer* problem, where the appearance of a clear image is transformed to match the visual style of a specific weather condition or **(b)** as a form of *weather augmentation*, where GANs are used to synthesize realistic weather effects to expand training datasets for robust model learning. Both approaches leverage the powerful image-to-image translation capabilities of GAN architectures to produce visually realistic and semantically consistent outputs under various weather conditions.

The paper [15] investigates the use of neural style transfer (NST) to simulate weather effects in clear-weather images. Building upon the method introduced by [16], the authors apply NST by jointly optimizing an input image to match the content features of a clear-weather image and the style features of a weather-degraded reference image. The process involves an iterative optimization scheme where an input image is modified to minimize two key objectives: content loss, which preserves the high-level structure of the original clear image, and style loss, which encourages the output to adopt the low-level texture statistics of a target weather-style image. These losses are computed from the feature maps of a convolutional neural network—in this case, a modified VGG19 network.

The authors adapt the process for weather-specific style transfer by fine-tuning the VGG19 model as a weather classifier. This fine-tuned network is used to better capture and represent weather-related texture features during the transfer process. This allows the style loss to be more sensitive to features relevant to rain streaks, fog diffusion, or snow granularity rather than artistic brushstrokes or color palettes. However, NST is computationally intensive, requiring multiple forward and backward passes through the network for each image. Additionally, it lacks explicit semantic control, which can lead to style inconsistencies across objects or distortions in critical regions, potentially hindering its use in downstream tasks.

Paper [17] does not directly address the task of adverse weather generation; it proposes a framework that is highly applicable to it by tackling a closely related problem, seasonal image translation. The authors demonstrate how their model can transform scenes between different seasons, such as from summer to winter, by learning the stylistic differences between domains while preserving the underlying structure of the image. This seasonal variation shares many characteristics with adverse weather synthesis, as both involve altering the visual style of the environment (e.g., textures, tones, atmospheric effects) while maintaining semantic consistency.

Their proposed approach enables diverse and unpaired image-to-image translation by disentangling image representations into two parts: a domain-invariant content code and a domain-specific attribute (style) code. The content encoder captures shared structure across domains like spatial layout, while the attribute encoder captures the style unique to each domain (e.g., snow, fog, or autumn foliage). During training, images from different domains are encoded into their respective content and style representations. By swapping the style codes between the domains and generating new images, the model ensures that the resulting images maintain the original content while adapting to the new style. A cross-cycle consistency loss ensures that the content remains preserved after translation and re-encoding. Differing from traditional cycle-consistency loss, cross-cycle consistency is applied by re-encoding the generated images and swapping the original style codes back in, reconstructing the original domain. This loss encourages the final output to closely resemble the original input, effectively regularizing the disentangled space while preserving semantic structure across translations.

Building on the seasonal image translation framework discussed previously, the *Multimodal Unsupervised Image-to-Image Translation* (MUNIT) [18] model adopts a similar disentanglement strategy by separating images into content and style representations. However, it specifically addresses the challenge of unpaired data by designing a fully unsupervised framework capable of multimodal image-to-image translation. The architecture includes separate content and style encoders for both source and target domains. The content encoder captures domain-invariant structural information, while the style encoder learns domain-specific appearance attributes. By sampling different style codes from the target domain’s latent style space and combining them with the content code of a source image, the decoder can generate diverse and plausible outputs, each reflecting a different stylistic variation within the target domain.

To ensure the quality and coherence of the translated images, the authors incorporate several loss functions into the training process. The image reconstruction loss ensures that an image can be accurately reconstructed from its content and style codes, preserving the integrity of

the original image. The latent reconstruction loss enforces that the content and style codes can be recovered from a generated image, promoting consistency in the learned representations.

In addition to analytical augmentation and neural style transfer, the authors of [15] also employed a GAN-based approach to synthesize weather-degraded images from clear-weather inputs. Specifically, they adopt CycleGAN [9] to model the mapping between clear and adverse weather domains. The model consists of two generators and two discriminators, similar to [19]: one generator translates clear-weather images into weather-affected versions (e.g., fog, rain, snow), while the second performs the inverse. These weather-translated images are then used to augment object detection models during training to enhance robustness under challenging conditions. While the approach successfully produces realistic transformations, it occasionally introduces artifacts or distortions, particularly around object boundaries, which reduces detection accuracy.

Additionally, a few recent works that aimed to address the challenge of multi-weather image translation using a unified generative framework were WeatherGAN[3] and WeatherGAN-UNet[20]. Both models seek to move beyond pairwise domain translation, instead offering a single, scalable architecture capable of translating images across multiple weather conditions, such as sunny, rainy, foggy, and snowy. However, while they share a common objective, their architectural approaches and translation strategies differ. WeatherGAN[3] focuses on weather-cue awareness by explicitly segmenting and attending to weather-specific regions in the image. Its architecture is composed of three main modules: an initial translation module for global transformation, a weather-cue segmentation module to identify condition-relevant areas, and an attention module to selectively apply transformations to those regions. This structure allows the model to isolate and manipulate local weather cues, preserving background and scene content effectively.

In contrast, WeatherGAN-UNet[20] adopts a more unified design through a single UResNet generator, which merges U-Net's [21] spatial preservation capabilities with ResNet's [22] deep residual learning. Rather than explicitly segmenting weather cues, it uses cycle consistency and adversarial learning in an unsupervised setup to learn domain mappings implicitly, treating different weather types as target domains and encouraging content preservation through architectural and loss design. While WeatherGAN-UNet emphasizes spatial focus and region-level editing via attention, WeatherGAN relies on an end-to-end content-preserving generator to maintain semantic structure during translation. Each model has trade-offs: WeatherGAN-UNet offers finer control through cue segmentation but depends heavily on the accuracy of weather-cue identification, while WeatherGAN provides efficiency and simplicity through architectural unification but may struggle to express fine-grained weather attributes precisely.

While the models represented earlier focus solely on translating images between weather domains, the work presented in [23] addresses both the generation and removal of weather effects within a unified framework. The proposed system includes a generative model that creates images with adverse weather conditions from clear scenes, as well as a restoration network trained to recover clear images from adverse weather conditions. By pairing these two tasks, the framework not only supports data augmentation for training robust models but also provides an image enhancement mechanism that improves visual clarity during infer-

ence. The restoration model is trained using synthetically generated image pairs, allowing for supervised learning even in the absence of real paired weather data. Although the framework introduces an end-to-end solution with the ability to add and remove adverse weather conditions, it faces challenges related to the realism of synthetic data and the computational demands of jointly learning two tasks.

Across both analytical and learning-based approaches, the task of generating adverse weather conditions has largely been addressed through the lens of image-to-image translation, with an emphasis on visual realism and appearance-level transformation. Whether relying on physics-inspired simulations or deep generative models like GANs, most existing methods focus on modifying the global style or texture of an image to reflect different weather conditions. However, very few of these approaches incorporate semantic segmentation information during generation, which limits their ability to maintain fine-grained semantic boundaries and object consistency, especially in tasks like object detection. This lack of structure-aware translation can result in artifacts or semantic misalignments, which is the gap we aim to address in this research.

2.5 Semantic Segmentation Models

Semantic segmentation is a computer vision task that involves assigning a class label to each pixel in an image [24]. The goal is to segment the image into regions corresponding to specific object categories, such as roads, cars, buildings, or pedestrians. Unlike object detection, which identifies and localizes objects using bounding boxes, semantic segmentation provides a dense, pixel-level understanding of the scene. This fine-grained classification is essential for applications like autonomous driving, medical imaging, and scene parsing, which is the main reason we chose the semantic segmentation task as our downstream task to assess the semantic consistency and correctness of our generated images with adverse weather conditions.

Dilated Residual Networks (DRNs) [25], introduced by Fisher Yu and Vladlen Koltun, present an innovative approach to enhancing convolutional neural networks for semantic segmentation and object localization. Traditional convolutional networks often reduce the spatial resolution of feature maps through pooling and strided convolutions, which can hinder the model’s ability to capture fine-grained details essential for dense prediction tasks. DRNs address this limitation by integrating dilated (or *trous*) convolutions into residual network architectures, effectively expanding the receptive field without compromising spatial resolution. This design choice allows DRNs to maintain high-resolution feature maps, enabling more precise localization and segmentation of objects within an image. Architecturally, DRN builds on the residual learning framework of ResNet [22] by replacing some of the stride-based downsampling operations with dilated convolutions. This change enables the model to achieve a large receptive field while retaining the ability to precisely localize object boundaries—an essential characteristic for evaluating whether generated images respect the spatial structure imposed by segmentation labels. Furthermore, the authors identify and mitigate gridding artifacts, a common issue associated with dilated convolutions, by introducing a ‘degridding’ technique that refines feature representations and enhances model performance. Empirical evaluations demonstrate that DRNs outperform their non-dilated

counterparts in image classification tasks without increasing model depth or complexity. Notably, a 42-layer DRN surpasses a deeper ResNet-101 model on the Cityscapes dataset by over 4 percentage points in mean Intersection-over-Union (mIoU), underscoring the efficacy of DRNs in preserving spatial acuity and improving accuracy in dense prediction tasks.

3 Methodology

In this work, we approach the task of generating semantically consistent images under adverse weather conditions by dividing them into two components. The first component focuses on **semantic image synthesis**, where we employ a GAN model to generate foggy images conditioned not only on segmentation label images but also on corresponding adverse weather inputs. In addition, to further guide the generation process, we augment the GAN model with **paired clear-weather images** captured from the same locations. These clear images are used as references to inform the generator about the scene structure under normal conditions. In this section, we explore various strategies to incorporate this reference information into the semantic image synthesis model.

The second component addresses the **downstream task: semantic segmentation**. To this end, we train a dedicated segmentation model using a dataset composed of both real foggy images and synthetic ones generated by the GAN model. By comparing segmentation performance across different training distributions, e.g., real-only, synthetic-only, and real + synthetic, we aim to assess the impact of semantic guidance in the generation process. This setup allows us to empirically validate whether the inclusion of segmentation maps during image synthesis leads to outputs that preserve the class-wise structure. In essence, this two-part methodology not only targets high-quality image generation under adverse conditions but also evaluates the *semantic consistency* of the generated images.

3.1 Semantic Image Synthesis Model

After comparing semantic image synthesis models, such as the OASIS framework [12], CycleGAN [9], DualPyramid GAN [13], we ultimately selected the *DualPyramid GAN* (DP-GAN) for our methodology due to its higher performance on our task. Unlike more general-purpose models, DP-GAN was designed with a particular emphasis on balancing global scene structure and local detail, which is crucial when generating images for structured environments like urban streets. One key reason for choosing DP-GAN was its demonstrated resilience on small and imbalanced datasets. In contrast to heavier models that tend to overfit or fail to generalize under limited data conditions, DP-GAN maintains consistent performance by explicitly leveraging semantic priors through SPADE layers and reinforcing generation across multiple spatial scales. Its use of a dual-path generator, while more complex, proved to be more effective in retaining fine-grained semantic consistency, particularly in regions where fog tends to obscure object boundaries.

Moreover, unlike OASIS, which primarily focuses on class-aware supervision through its discriminator, DP-GAN extends this idea by enhancing the generation side as well, ensuring both adversarial and generative components are semantically aligned.

3.1.1 Clear Weather Input Incorporation

Initially, the DP-GAN model was trained using only semantic segmentation maps as input to generate images with fog conditions. While this setup provided the generator with a structured understanding of the scene layout, such as object positions and class information, it lacked critical contextual cues (e.g., time of day, traffic flow) and visual cues (e.g., textures, shadows, reflections). As a result, the generated images often appeared oversimplified with limited detail. This highlighted a fundamental limitation: the segmentation maps alone were not enough to guide the generation process toward producing contextually and visually rich adverse weather images.

To address these shortcomings, we extended the input to include the corresponding clear weather image as a reference. This additional input provides the model with a rich source of semantic and appearance-level information, helping to preserve scene geometry and fine-grained details. By conditioning the generation process on a clear-weather reference, the GAN is better able to model the transformation from clear to adverse conditions, rather than attempting to synthesize an entire scene from abstract labels alone.

This approach draws inspiration from image-to-image translation frameworks such as *Pix2Pix* and *CycleGAN*, which demonstrate the benefits of learning mappings between paired domains. In our case, the clear weather image serves as a grounded starting point where the GAN can learn how specific visual features are affected by adverse conditions. This not only improves the realism and consistency of the outputs but also reduces ambiguity during training.

Moreover, adding the clear weather reference enhances the model’s robustness and diversity by reducing the likelihood of mode collapse. When used in conjunction with semantic segmentation maps, this setup creates a multi-modal conditioning framework: the segmentation map guides the structural layout of the scene, while the clear image provides fine texture and appearance details. The following section provides a detailed explanation of the technique used to incorporate clear weather images into the generation process.

Three-Pyramid Generator

In our proposed method, *3P-WeatherGAN*, we introduce a third pyramid to the DP-GAN generator to specifically process the clear weather image. While the original Dual Pyramid GAN architecture uses two pyramids (typically for encoding and decoding feature hierarchies of semantic labels and adversarial feedback), this new third pyramid serves as a dedicated encoder or feature extractor for the clear weather reference.

As shown in Figure 3.1, the new generator works as follows:

- **Pyramid 1:** Processes the semantic segmentation map, extracting multi-scale structural features.
- **Pyramid 2:** Performs multi-scale synthesis and reconstruction to generate the adverse weather image.
- **Pyramid 3 (new):** Processes the clear weather image through its encoder path, extracting fine-grained appearance features at multiple scales.

Through this architecture, the third pyramid independently processes the clear image through a sequence of convolutional and downsampling layers similar to the semantic label pyramid (pyramid 2). This new pyramid acts as a feature pyramid that captures global scene layout, mid-level structure, and fine-grained textures at multiple resolutions. The new generator model will now receive the following:

- A semantic segmentation map (optionally concatenated with 3D noise), which is processed through the segmentation pyramid.
- A clear weather reference image, which is passed through the clear weather pyramid.

Three-Pyramid Generator: Dual SPADE Normalization

In this strategy, we extend the original SPADE-based conditioning mechanism to incorporate **two distinct SPADE normalization blocks** at each decoding stage. One SPADE block is conditioned on the *semantic segmentation pyramid*, while the other is conditioned on the *clear weather feature pyramid*. This *dual-spade normalization mechanism* enables the generator to separately process structural and appearance guidance before integrating them into the decoding process.

In the standard DP-GAN generator architecture, each decoder layer is modulated by a single SPADE block conditioned on semantic features. However, in our setting, the availability of a clear-weather reference image offers an additional source of context information.

As shown in Figure 3.1, at every resolution level in the decoder, the intermediate feature map is sent through:

- A SPADE block modulated by the segmentation feature map extracted from the **semantic encoder pyramid**.
- A SPADE block modulated by the visual feature map extracted from the **clear weather encoder pyramid**.

These two SPADE paths produce *two distinct normalized versions* of the same decoder feature map. This design maintains a clean separation of modalities, allowing each conditioning stream to inject its own spatially adaptive transformations. Through this setup, the model does not conflate semantic layout and photorealistic detail at the conditioning stage but instead allows them to independently influence the output generation process.

Once both modulated outputs are obtained, they are **combined through a separate fusion mechanism**, namely *parallel fusion* (with learnable weighted blending), which is described in the following section.

Parallel Fusion:

For this fusion, the outputs of the two SPADE blocks are fused through a learnable interpolation mechanism. After applying separate SPADE normalization to the decoder feature map using both the segmentation and clear weather conditioning inputs, the resulting tensors are linearly combined using a learned scalar parameter α . This parameter is optimized during training and determines the relative weighting of each conditioning stream.

The fused tensor is computed as:

$$\text{Fused} = \alpha \cdot \text{SPADE}_{\text{seg}}(x) + (1 - \alpha) \cdot \text{SPADE}_{\text{clear}}(x), \quad (3.1)$$

where $\alpha \in [0, 1]$ after applying a sigmoid activation to ensure interpretability and numerical stability.

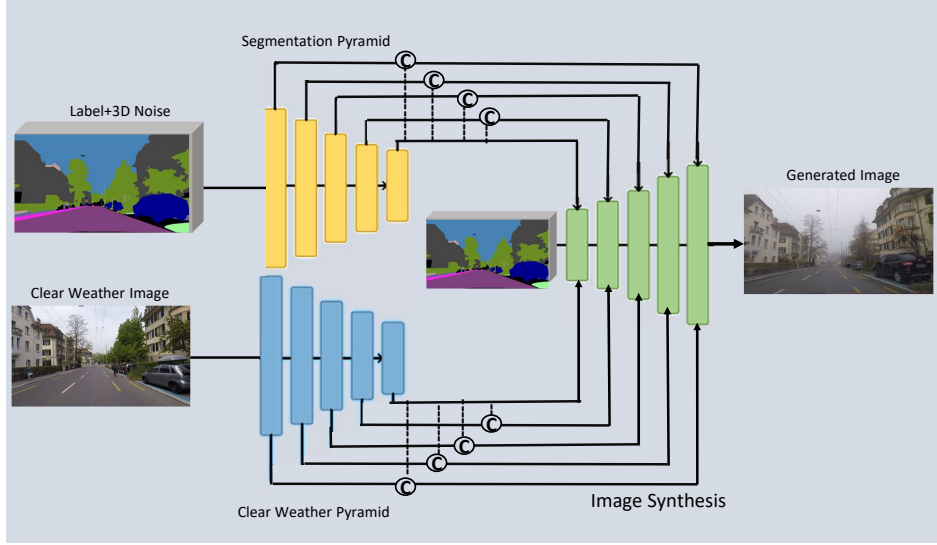


Figure 3.1: 3P-WeatherGAN: The generator is guided by both semantic layout from the segmentation pyramid (yellow blocks) and realistic appearance from clear weather image pyramid (blue blocks) across all spatial scales of the image synthesis pyramid (green blocks) through two **separate SPADE** blocks. Figure adapted from [13]

When α approaches 1, the generator relies more heavily on semantic information; when it approaches 0, clear-weather features exert greater influence. This provides the generator with the flexibility to selectively prioritize one modality over the other depending on the context of the generation task.

3.1.2 Loss Functions for Clear Weather Image Guidance

To fully exploit the additional supervisory signal provided by the clear weather reference image, we introduce two new loss components to the generator’s objective: a feature-level alignment loss and a perceptual loss, both computed with respect to the clear input image. These losses are designed to ensure that the generated output is not only structurally consistent with the semantic layout but also visually aligned with the reference appearance under clear conditions.

Feature Alignment Loss with Clear Weather Image

The first newly introduced term is a feature alignment loss between the generated image and the clear weather reference image. To compute this, both the synthesized image and the clear image are passed through the discriminator network, and their intermediate feature representations are extracted. Denoting the extracted features as f_{fake} for the generated image and f_{clear} for the clear input, we define the alignment loss as:

$$\mathcal{L}_{\text{align-clear}} = \sum_i \|f_{\text{fake}}^i - f_{\text{clear}}^i\|_1,$$

Where the sum runs over selected layers of the discriminator. This loss encourages the internal representations of the fake and clear images to be similar, thereby guiding the generator to embed visual patterns that are consistent with the clear scene structure and content.

VGG Perceptual Loss with Clear Weather Image

The second loss component added is a VGG-based perceptual loss that directly measures the appearance similarity between the generated image and the clear weather reference. This loss is defined as:

$$\mathcal{L}_{\text{VGG-clear}} = \sum_j \frac{1}{C_j H_j W_j} \|\phi_j(G(x)) - \phi_j(I_{\text{clear}})\|_1,$$

Where $\phi_j(\cdot)$ denotes the activation of the j -th layer in a pretrained VGG-19 network, and C_j, H_j, W_j are the channel, height, and width dimensions of the corresponding feature map. $G(x)$ denotes the generator output, and I_{clear} is the clear reference image.

This loss encourages the generated image to preserve high-level appearance features, such as texture, object surface properties, and lighting characteristics, that are present in the clear image.

3.2 Downstream task: Semantic Segmentation

The second component of our methodology is aimed at evaluating the practical utility of the generated adverse weather images in a downstream task — semantic segmentation. While visual quality and perceptual realism are important, it is equally critical to assess whether the synthesized images are semantically meaningful and structurally coherent. To this end, we adopted the **Dilated Residual Network (DRN)** as our backbone segmentation model.

This setup rigorously evaluates the semantic consistency and structural fidelity of our image synthesis methods. It also highlights the potential of our framework to augment datasets for challenging weather conditions, where labeled data is scarce or expensive to acquire.

4 Evaluation

4.1 Datasets

In this section, we will review commonly used datasets for adverse weather image generation, outlining their relevance and limitations. Additionally, we present the dataset that was ultimately used, highlighting the specific factors that made it more suitable for achieving the objectives of our research.

Several publicly available datasets have been considered for our task; however, each presented limitations that made them unsuitable for this task. The Boreas dataset [26], while highly relevant due to its focus on challenging environmental conditions such as snow, rain, and fog, lacks dense semantic annotations. This absence of ground truth labels makes it less suitable for training or evaluating image translation models that require detailed scene understanding.

On the other hand, datasets like BDD100K [27] and ADE20K [28] do provide semantic annotations, yet they also come with critical limitations. BDD100K includes rich weather metadata and high-quality urban street scenes under various conditions, including nighttime and rainy weather, making it a strong candidate in terms of content. However, the dataset does not offer paired images of the same scene under different weather conditions, which is needed for our conditional image-to-image translation task. Moreover, adverse weather conditions like fog and snow are underrepresented. ADE20K, although densely annotated with pixel-level semantic segmentation for a wide variety of scenes, was designed for general scene parsing and not for driving scenarios. As such, it lacks a consistent urban or street-level focus and does not include any views of the same scene under varying conditions.

Overall, these datasets either lacked necessary annotations or scene consistency across weather states, motivating the selection of an alternative dataset.

The Cityscapes dataset [29] was a strong candidate for our task due to its dedicated focus on urban street scenes and its provision of high-quality semantic annotations. It contains 5,000 finely annotated images, with pixel-level segmentation across 30 visual classes (e.g., roads, buildings, cars, pedestrians), all captured from vehicle-mounted cameras in multiple European cities. This rich annotation makes it highly suitable for semantic image synthesis and domain adaptation. The dataset strikes a strong balance between data volume and label quality, offering enough diversity to support robust model training while remaining computationally manageable. However, the dataset is limited to clear weather conditions and daytime scenes, lacking the environmental diversity needed for learning transformations across different weather domains.

The previously mentioned shortcomings motivated the use of *The adverse conditions dataset with correspondences (ACDC)* dataset [30]. The ACDC is a high-quality semantic segmen-

tation dataset specifically designed to support research in adverse weather and lighting conditions in urban driving environments.

The dataset consists of **4,006 finely annotated images**, all captured in real-world urban environments from vehicle-mounted cameras in multiple urban settings with a focus on European cities. Each image is labeled with *dense pixel-level semantic annotations* following a class structure closely aligned with the Cityscapes taxonomy, making it compatible with models pre-trained on Cityscapes and ideal for transfer learning.

ACDC is organized into four distinct weather condition subsets:

- **Clear** (normal daytime conditions)
- **Fog**
- **Night**
- **Rain**

Each subset contains approximately **1,000 images**. All scenes are real (not synthetic) and were carefully selected to ensure diversity in terms of city layout, traffic density, and scene composition.

Key Structural Features of ACDC dataset:

- **Semantic Labels:** Each pixel in the images is assigned one of 19 classes, consistent with the Cityscapes taxonomy (e.g., road, sidewalk, building, car, pedestrian).
- **Scene Consistency and Paired Domains:** ACDC features *pairing* across weather domains. Many scenes were captured at the **same geographic locations** under different weather conditions (e.g., both clear and foggy), providing approximate pairing for domain translation tasks. However, the pairing is not perfect—dynamic elements such as cars, pedestrians, and vegetation may differ between conditions due to the variation of the natural scene at different times.
- **High Resolution:** Images are provided at a resolution of 1920×1080.

The ACDC dataset provides a variety of annotation types, including standard semantic segmentation labels, panoptic segmentation, instance-level labels, and invalid region masks. In this work, we exclusively use the standard semantic segmentation annotations, which form the basis for the summary presented in Table 4.1. The table provides an overview of the structure of the ACDC dataset in the three adverse weather domains: *Fog*, *Rain*, and *Night*. For each domain, the total number of images is distributed among 400 training images, 100 validation images, and 100 test images. All adverse-weather images are accompanied by semantic segmentation annotations (*Annotated Images* column) except for the test set. Additionally, each image in these domains has a corresponding clear weather image captured from a similar location, enabling approximate domain pairing (*Paired Clear Images* column). However, only half of these clear-weather images include matching semantic segmentation annotations (*Paired Clear with Annotations* column), limiting the number of fully annotated pairs available for supervised learning to 200 training images and 50 validation images per domain. Figure 4.1 illustrates a sample from the foggy domain of the ACDC dataset used in our experiments.



Figure 4.1: ACDC dataset sample: Each foggy image (a) is accompanied by a corresponding semantic annotation (b), as well as a paired clear-weather image (c) captured from the same location. The clear-weather image is also provided with its own semantic annotation (d).

Table 4.1: Summary of ACDC dataset structure across adverse weather domains. Only standard semantic annotations are counted.

Domain	Total Images	Annotated Images	Paired Clear Images	Paired Clear with Annotations
Fog	600	600	600	250
Rain	600	600	600	250
Night	600	600	600	250

4.2 Experiments

This section presents the experimental setup conducted to assess the performance of both the baseline and our proposed methodology (*3P-WeatherGAN*). First, we describe the setup of baseline experiments using well-established state-of-the-art models for image-to-image translation, such as OASIS and Dual-pyramid GAN, applied to the ACDC dataset. These serve as reference points for performance comparison. Following that, we detail the experiments conducted using our proposed methodology, including the training configurations and implementation details.

4.2.1 Baseline Experiments

In the baseline experiments, two distinct training directions were explored to evaluate different model paradigms for our task:

- The first direction involved training a classic **unpaired image-to-image translation model** CycleGAN [9], which does not make use of semantic segmentation maps. Instead, CycleGAN learns to translate between domains only by the cycle-consistency loss. This setup was intended to assess how well a model can learn meaningful features and perform domain translation relying solely on visual cues from real images, without any semantic guidance.

- The second direction focused on **semantic image synthesis** models that explicitly condition the generation process on segmentation labels. In this setting, two state-of-the-art models were employed: **OASIS** [12] and **DP-GAN** [13].

For the first baseline experiment, **CycleGAN** was trained to perform unpaired image-to-image translation from the clear to the foggy domain. The model was trained for **100 epochs** with a **batch size of 2**, using the **Adam optimizer** with a learning rate of **0.0002** and default momentum parameters ($\beta_1 = 0.5, \beta_2 = 0.999$). The training was conducted on a server equipped with **NVIDIA GTX 1080 Ti GPU 11GB VRAM**. Input images were resized to a width of 512 pixels while maintaining their original aspect ratio, followed by normalization as per standard CycleGAN preprocessing. The training process followed the original CycleGAN framework represented in [9] without architectural modifications.

For the second baseline, the **OASIS** framework was trained on the fog domain as well, but using the provided semantic segmentation images as input. The model was trained for **400 epochs** with a **batch size of 5** on a server equipped with an **NVIDIA RTX A5000 GPU 24 GB VRAM**. OASIS employs a fully convolutional generator that consists of a series of residual blocks with upsampling layers, where the input semantic label map is progressively transformed into an image through learned feature representations. The object-aware feedback loop enables OASIS to generate high-resolution, semantically accurate, and visually consistent images, even for complex scenes with diverse object categories. In the context of this research, integrating ideas from OASIS allows for more controlled and detailed weather condition synthesis, ensuring that added fog effects, for example, align naturally with the scene structure and object types.

During training, the **VGG perceptual loss** was enabled, which leverages feature activations from a pretrained VGG network to measure high-level perceptual similarity between generated and real images. This loss plays an important role in enhancing texture realism and preserving the semantic integrity of generated outputs. The training followed the official OASIS implementation represented in [12] without architectural modifications.

The third and final baseline experiment involved training the **DualPyramid GAN (DP-GAN)** model [13] on the fog domain of the ACDC dataset using semantic segmentation maps as input. The training was conducted for **400 epochs** with a **batch size of 5** on an **NVIDIA RTX A5000 GPU (24 GB VRAM)**. The **VGG perceptual loss** was also enabled during training to improve texture realism and perceptual quality.

All models were trained from scratch to evaluate their capacity to learn from a limited dataset—in this case, the fog domain of the ACDC dataset, consisting of 400 training images and 100 validation images. Notably, in this phase of experimentation, neither OASIS nor DP-GAN models were exposed to the corresponding clear-weather images. This was done intentionally to assess their performance in generating realistic foggy images based solely on semantic inputs, without paired supervision.

To ensure a fair comparison between all baseline models, the same training environment, dataset split, and preprocessing pipeline were used across all experiments. Notably, the data loading scripts for both OASIS and DP-GAN were originally designed for the Cityscapes dataset, which has a highly similar structure and annotation format to ACDC. As a result,

minimal modifications were needed to adapt the pipelines, like folder structure, and no additional resizing, label remapping, or custom transformations were required beyond the default preprocessing. This helped streamline integration and reduce preprocessing bias, ensuring that both models were evaluated under equivalent conditions.

4.2.2 Proposed Methodology Experiments

Semantic Image Synthesis Model

To evaluate the impact of incorporating clear-weather reference images using the three-pyramid approach, we proposed (*3P-WeatherGAN*). We trained our model using a **single 24GB GPU, with a batch size of 2, trained for 500 epochs**. The training setup utilized VGG loss in addition to a feature alignment loss designed to preserve consistency with real-weather images as described in 3.1.2.

The training data consisted of the 400 fog-domain images from the ACDC dataset, along with their corresponding clear-weather images and segmentation annotations. The annotations used for input were those corresponding to the foggy images to maximize the usable training set size. Although each foggy image is paired with an annotation image, this is not always true for the clear weather images. Using the annotations from the clear-weather images would reduce the training set to only **250 usable samples**. However, the segmentation maps are largely consistent across both domains thanks to the weak-pairing property of the ACDC dataset. This approach allows us to retain more data from the fog domain, ensuring better model generalization while still leveraging the clear-weather images for their unobstructed scene structure and texture guidance. For validation, our goal was to simulate real-world deployment of the generator, where only the clear-weather image and its corresponding annotation would be available. Thus, the **50 validation samples** were selected to include clear-weather images with exactly matching fog-domain counterparts and annotations. This setup allows for a realistic and controlled evaluation, enabling comparison between the synthesized foggy images and their real fog-domain equivalents.

Semantic Segmentation Model

In these experiments, the synthetic foggy images were used as data augmentation alongside the real fog domain images from the ACDC dataset. The goal was to evaluate whether augmenting the training set with additional GAN-generated samples could improve the downstream semantic segmentation performance. The synthetic images were generated using two distinct methods: our proposed **3P-WeatherGAN** framework and the baseline CycleGAN model. By incrementally increasing the number of synthetic images during training, we assessed the added value of each generation technique in enhancing segmentation accuracy in the fog domain. For training the *Dilated Residual Network* (DRN), we used the DRN-D-22 architecture. The evaluation was performed using the standard 19-class label set, which is compatible with Cityscapes taxonomy. The DRN model was trained for **100 epochs** using **stochastic gradient descent (SGD)** with a **learning rate of 0.01** scheduled to decay at **epoch 50, momentum of 0.9**, and a learning rate decay applied at epoch 50. The input images were resized to a fixed height of 720 pixels, and a **batch size of 5** was used on a server equipped

with an **NVIDIA RTX A5000 GPU 24 GB VRAM**. To improve generalization, data augmentation techniques such as random rotation (± 10 degrees) and random scaling ($\times 2$) were applied. Each experimental setup maintained the same training procedure, differing only in the proportion of synthetic images added to the real fog-domain training set.

This setup allowed us to assess whether unpaired translation methods, in this case, CycleGAN, could serve as effective alternatives to our structured, segmentation-guided image synthesis (3P-WeatherGAN) in terms of improving downstream segmentation performance under foggy conditions.

4.2.3 Ablation Study

In this section, we explore and evaluate several techniques to incorporate clear-weather reference images into the generative framework to effectively guide the image synthesis process. These experiments were essential in understanding how to best leverage the structural and textural cues present in the clear-weather images. We investigated multiple integration strategies, including concatenating the reference image at the input level, feature-level fusion through shared encoder layers, and dual-path conditioning using SPADE blocks. Each variant was assessed based on its impact on generation quality, realism, and semantic fidelity.

Channel Concatenation

As a starting point, we explored the use of direct channel concatenation of the clear weather image and its corresponding semantic segmentation image as input to the DP-GAN model. As shown in Figure 4.2, we constructed the input by stacking the clear weather RGB image with the one-hot encoded segmentation map along the channel dimension, forming a single, fused tensor. This composite input allows the generator to simultaneously access both appearance-level information from the clear image and structural guidance from the segmentation map.

Formally, given a clear weather image $I_c \in \mathbb{R}^{H \times W \times 3}$ and a segmentation map $S \in \mathbb{R}^{H \times W \times k}$, where k is the number of semantic classes (in our case 34 classes), we concatenate them to form an input tensor $X \in \mathbb{R}^{H \times W \times (3+k)}$ making the final input 37 classes. This fused tensor is then passed as input to the generator network of the DP-GAN. The key advantage of this design is that it enables the early fusion of both modalities, allowing the network to learn joint representations of texture, object classes, and spatial layout from the very first convolutional layer.

This setup fits naturally within the architecture of the Dual Pyramid GAN, which is designed to process and refine features through multiple scales. Because the concatenated input preserves perfect spatial alignment between the clear image and segmentation map, it provides a coherent and information-rich input at all pyramid levels.

Moreover, this approach offers a practical benefit in terms of architectural simplicity. By treating the clear image and segmentation map as a single input stream, we avoid the need for separate encoders or complex fusion modules. This reduces computational overhead while still leveraging the strengths of both input types.

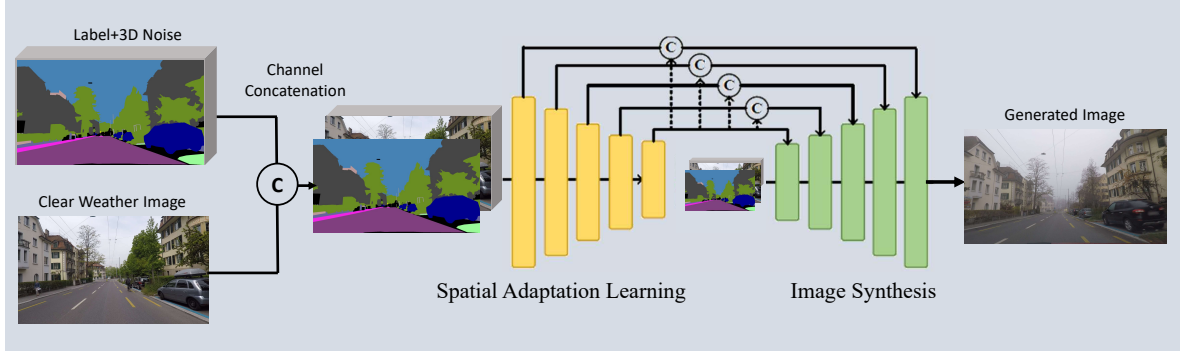


Figure 4.2: DP-GAN generator architecture after clear input fusion: the generator is passed as input the segmentation label with the random noise (*label+3D noise*) **concatenated** with the matching *clear weather image*. Figure adapted from [13]

Three-Pyramid Generator: Feature Fusion Techniques

In contrast to direct concatenation, which fuses information at the pixel level before feature extraction, the three-pyramid approach allows the clear image to be encoded through its hierarchical pathway, learning multi-level features that are semantically aligned but distinct from those derived from the segmentation map. We experimented with two strategies to combine the output features of the third pyramid (clear image) with those of the segmentation map pyramid. The goal of these integration strategies is to provide the synthesis pathway with a contextually rich and visually accurate representation.

The integration strategies explored include the following:

1. Single SPADE normalization

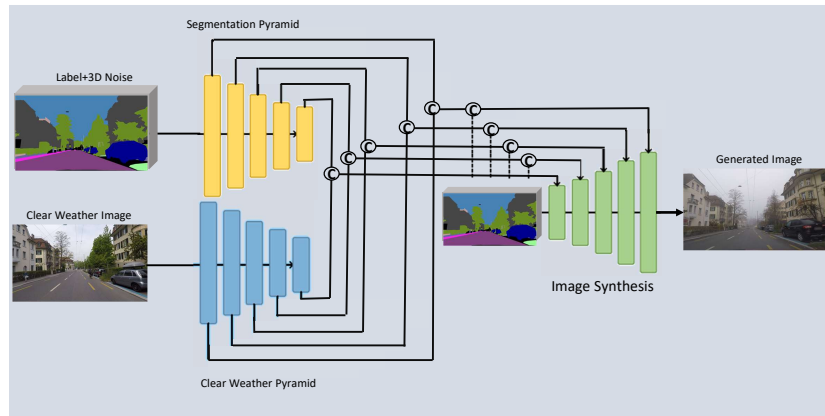


Figure 4.3: Single SPADE three pyramid generator Architecture. The generator is guided by both semantic layout from the segmentation pyramid (yellow blocks) and realistic appearance from the clear weather image pyramid (blue blocks) across all spatial scales of the image synthesis pyramid (green blocks) through feature map concatenation. Figure adapted from [13].

In this strategy, the feature maps produced by the semantic segmentation and clear image pyramids are concatenated and then fed jointly into the SPADE-modulated ResNet (decoder)

blocks within the generator. This enables the model to condition the spatially-adaptive normalization using combined context from both semantic and appearance domains, while still leveraging the standard SPADE mechanism to maintain spatial awareness. Figure 4.3 illustrates the new generator architecture used:

At each resolution level, the outputs of the two pyramids are concatenated. This is represented in the figure by the circular nodes labeled "C". These concatenated features are then passed to the decoder (green blocks), where they serve as conditioning inputs to the SPADE-modulated ResNet blocks.

Importantly, the decoder feature map at each stage is modulated using a **single SPADE normalization block**. This design enables SPADE to spatially modulate the decoder features based on a joint representation of the scene structure and clear-weather appearance. By fusing the feature maps before SPADE normalization, this method benefits from rich cross-modal information while maintaining the original spatial adaptation flow.

2. Dual SPADE Normalization

As previously mentioned, the two SPADE paths produce *two distinct normalized versions* of the same decoder feature map. This design maintains a clean separation of modalities, allowing each conditioning stream to inject its own spatially adaptive transformations. Through this setup, the model does not conflate semantic layout and photorealistic detail at the conditioning stage but instead allows them to independently influence the output generation process. Once both modulated outputs are obtained, they are combined through a separate fusion mechanism, we explored sequential SPADE normalization. For this fusion, we adopt a **sequential SPADE** normalization strategy in which each decoder feature map is modulated first by the semantic segmentation features and then by the clear weather features. This approach differs from our parallel dual-SPADE method described in Section 3.1.1, in that the conditioning is applied in a *cascaded manner* rather than independently and fused post-modulation. As a result, this configuration enables the generator to apply a form of *hierarchical modulation*, where one modality influences the intermediate representation before the second modality is applied.

At each resolution level in the decoder, the intermediate feature map is first normalized using a SPADE block conditioned on the output of the semantic segmentation pyramid. This first modulation enforces structural coherence and spatial consistency with respect to the scene layout. The resulting normalized tensor is then passed through a second SPADE block, this time conditioned on the corresponding features from the clear weather pyramid. This second stage injects fine-grained appearance details, allowing the generator to refine the already-structured representation with photorealistic textures, lighting, and color cues present in the clear reference image.

By applying semantic conditioning before appearance conditioning, the generator first constructs a scene that is spatially and semantically coherent and then overlays that structure with realistic visual attributes.

Through this iterative process, we ultimately identified the three-pyramid architecture with dual SPADE normalization as the most effective and balanced approach for our task.

4.3 Evaluation Metrics

In evaluating the performance of the baseline models and the proposed methodology, we focused on two complementary directions: assessing the **visual realism** of the generated images and evaluating their **semantic correctness** concerning the input segmentation maps. To measure realism, we relied on quantitative metrics for GAN models. For semantic correctness, we conducted a secondary evaluation by training a dedicated segmentation model on datasets composed of both real and generated images, with varying proportions. This setup allowed us to test whether the generated images preserved sufficient semantic structure to contribute meaningfully to the segmentation task. For each experiment, we computed quantitative metrics on a validation set to evaluate how the inclusion of generated data affected the segmentation model’s performance.

Semantic Image Synthesis Model

To evaluate the visual realism of the generated images, we used the **Fréchet Inception Distance (FID)** as introduced in [31] and similarly used in [10, 32] for assessing the quality of images generated by GAN models. FID measures the distance between the feature distributions of real and generated images using activations extracted from a pre-trained Inception-v3 network. Specifically, it assumes that these features follow a multivariate Gaussian distribution and computes the Fréchet distance between the two.

The FID is computed as:

$$\text{FID}(X, Y) = \|\mu_X - \mu_Y\|_2^2 + \text{Tr}(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{\frac{1}{2}}) \quad (4.1)$$

Where μ_X, Σ_X and μ_Y, Σ_Y are the means and covariances of the Inception features of the real and generated images, respectively.

- μ_X, μ_Y : Mean vectors of the Inception feature representations for real and generated images.
- Σ_X, Σ_Y : Covariance matrices of the feature distributions.
- $\|\mu_X - \mu_Y\|_2^2$: Squared Euclidean distance between the means; measures average feature difference.
- $\text{Tr}(\cdot)$: Trace operator, i.e., the sum of diagonal elements of the matrix.
- $(\Sigma_X \Sigma_Y)^{1/2}$: Matrix square root of the product of the covariances; reflects how similarly the feature distributions vary.

The lower the FID score, the closer the distribution of generated images is to that of real images, indicating higher visual fidelity and realism. FID is preferred over earlier alternatives such as Inception Score because it compares both the *mean* and *covariance* of the feature embeddings, providing a more comprehensive assessment of the visual quality.

For a fair comparison, we computed the FID scores after running inference using the **best checkpoints** obtained at the end of training for each model. Inference was performed on the **ACDC fog-domain test set**, ensuring consistent and unbiased evaluation across all experiments.

Semantic Segmentation Model

The key metric used for evaluating the semantic segmentation model was the mean Intersection-over-Union (mIoU), a widely adopted metric in semantic segmentation that quantifies the overlap between predicted and ground truth regions across all semantic classes [33]. By computing the mIoU between the segmentation results produced by the DRN model and the original semantic labels used to guide generation, we assess how well the generated images preserve the class-level spatial structure. Higher mIoU scores indicate that the synthetic images maintain stronger semantic fidelity to the input layout.

To compute mIoU, we begin by evaluating the Intersection-over-Union (IoU) for each class. For a given class k , IoU is defined as:

$$\text{IoU}_k = \frac{TP_k}{TP_k + FP_k + FN_k}, \quad (4.2)$$

Where TP_k is the number of true positive pixels correctly predicted as class k , FP_k is the number of false positive pixels incorrectly predicted as class k , and FN_k is the number of false negative pixels belonging to class k but predicted as a different class.

The mIoU is then computed by averaging the IoU values across all K semantic classes:

$$\text{mIoU} = \frac{1}{K} \sum_{k=1}^K \text{IoU}_k. \quad (4.3)$$

5 Results

5.1 Baseline vs 3P-WeatherGAN Results

We assessed the three baseline models’ (i.e., CycleGAN, OASIS, DP-GAN) performance compared to our proposed methodology using the FID metric as shown in Table 5.1. The evaluation revealed that **DP-GAN outperformed OASIS** by achieving a lower FID score. Based on this result, we selected DP-GAN as the foundation for our proposed methodology. Among the baseline models tested, our proposed **3P-WeatherGAN** significantly outperforms the baselines, achieving the lowest FID score of **98.01**. This improvement is a direct reflection of the innovations we introduced in our architecture.

Unlike CycleGAN, which lacks semantic guidance, our method integrates semantic segmentation maps during training, ensuring that object boundaries and scene layout are maintained. Furthermore, compared to OASIS, which relies solely on segmentation input without explicit reference to actual clear-weather textures, our approach introduces a third pyramid to extract multi-resolution features from real clear-weather images. This not only helps the generator better preserve structural and textural fidelity but also reduces ambiguity in fog simulation, which is particularly challenging due to its inherently non-uniform nature.

The use of dual parallel SPADE blocks with a learnable fusion mechanism also plays a crucial role. This allows the model to dynamically balance contributions from both semantic structure (segmentation) and visual appearance (clear-weather input), resulting in more context-aware and photorealistic outputs.

Table 5.1: Comparison of baseline models against our proposed methodology. FID scores were computed between generated images and real fog-domain images from the ACDC validation set.

Baseline Model	FID
OASIS	122.1
CycleGAN	114.03
Dual-Pyramid GAN	111.2
3P-WeatherGAN (Ours)	98.01

A visual comparison of the generated outputs is presented in Figure 5.1, further illustrating the effectiveness of our approach. Compared to OASIS, our method produces images that are not only more realistic but also more semantically consistent, OASIS fails to preserve fine object boundaries and structural coherence. While CycleGAN sometimes achieves better alignment with segmentation structure, it lacks the dense fog realism and nuanced weather

effects that our model successfully captures. This balance between visual fidelity and semantic accuracy demonstrates the strength of our approach in synthesizing adverse weather conditions for downstream tasks.



Figure 5.1: Visual comparison of fog synthesis results using baseline models (a) OASIS and (b) CycleGAN against (c) our proposed methodology (**3P-WeatherGAN**) .

5.2 Ablation Study Results

Table 5.2 presents the FID scores obtained from the different fusion strategies explored in the methodology for integrating clear-weather reference images into the DP-GAN framework. Among the four experiments, the **3P-WeatherGAN** approach, which directly concatenates the clear-weather image with the segmentation map at the input level, achieved the lowest FID score of **98.01**, indicating the highest visual realism. Qualitative comparisons of generated foggy images using different fusion strategies are shown in Figure 5.2.

Table 5.2: FID scores for different fusion strategies used to integrate clear-weather reference images into the DP-GAN model.

Experiment	FID
Exp-PF3 (Pyramid Fusion - Dual Sequential SPADE)	111.5
Exp-PF1 (Pyramid Fusion - Single SPADE)	108.21
Exp-CF (Channel Fusion)	99.07
3P-WeatherGAN (Ours)	98.01

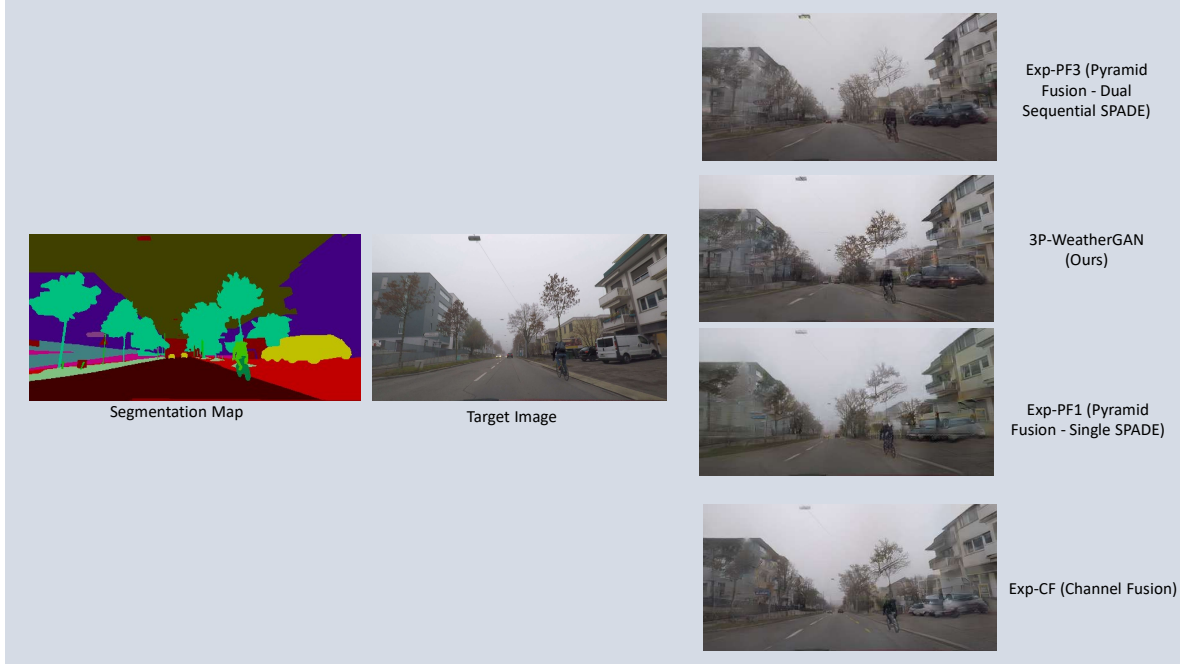


Figure 5.2: Visual comparison of fog synthesis results using different fusion strategies within our proposed methodology.

Based on this evaluation, we adopted the 3P-WeatherGAN approach as the final design for subsequent experiments, including the downstream segmentation task.

5.3 Segmentation Model Results

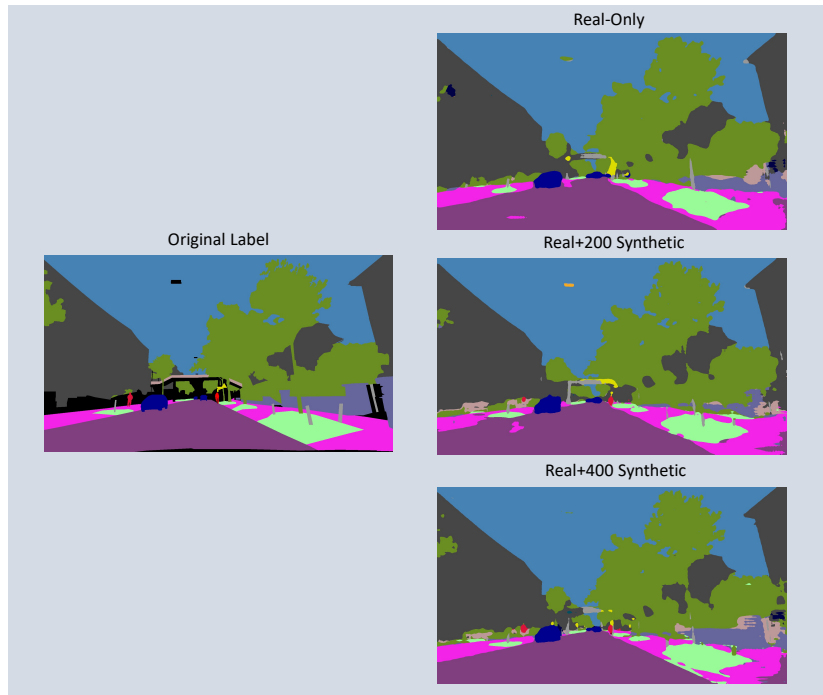
To evaluate the real-world utility of our generated synthetic foggy images, we conducted downstream task experiments focused on semantic segmentation. Specifically, we trained the DRN-C-26 model using an augmented dataset comprising both real foggy images and GAN-generated synthetic fog images from the ACDC dataset using various ratios of real and synthetic images.

The results shown in Table 5.3 reveal several key insights. First, adding synthetic data improves segmentation performance compared to using only real fog images **36.3** mIoU. With 200 synthetic images, our method outperformed CycleGAN **41.42 vs. 38.12**, indicating a more semantically useful generation. However, as the amount of synthetic data increased to 400 images, both methods converged to similar performance levels (42.05 for ours and 42.07 for CycleGAN). This suggests that while our model initially offers stronger guidance, the marginal gain from additional synthetic images may diminish due to saturation effects or domain redundancy. Overall, the results validate the effectiveness of our method in enhancing downstream performance, especially at lower synthetic-to-real ratios.

Table 5.3: Comparison of mIoU scores on ACDC Fog validation set using varying portions of synthetic images generated by CycleGAN and our proposed method

Training Dataset	Data Augmentation Method	mIoU
Real Fog Only		36.3
Real Fog + 200 synthetic images	CycleGAN	38.12
	3P-WeatherGAN (Ours)	41.42
Real Fog + 400 synthetic images	CycleGAN	42.07
	3P-WeatherGAN (Ours)	42.05

Figure 5.3 presents qualitative segmentation results for the models trained with increasing amounts of synthetic fog-domain data. As shown, the segmentation output becomes progressively more accurate and visually aligned with the original labels as the ratio of synthetic images increases.

Figure 5.3: Visualization of DRN model results trained on various proportions of synthetic images generated by our proposed methodology (**3P-WeatherGAN**).

5.4 Discussion

Despite these targeted improvements, our proposed model (**3P-WeatherGAN**), trained directly on ACDC, outperformed all other configurations with an FID of **98.01**, surpassing even the baseline models. Figure 5.4 presents results comparing original clear-weather images from the ACDC dataset with the outputs generated by our proposed **3P-WeatherGAN** model. It can be observed that the model successfully synthesizes fog effects with realistic texture and density while preserving scene structure and object layout. Fine details such as road markings, vehicles, and pedestrians remain clear, demonstrating that our approach is effective in introducing adverse weather features without compromising semantic consistency. This visual coherence highlights the model’s strength in generating plausible foggy conditions while retaining essential scene information needed for downstream segmentation tasks.



Figure 5.4: Visual results comparing original clear-weather images from the ACDC dataset (left column) with the outputs generated by our proposed **3P-WeatherGAN** model (right column).

An important feature of our proposed **3P-WeatherGAN** model is its ability to offer scene-level control through semantic guidance. By conditioning generation on both a segmentation label map and a clear-weather reference image, the model enables direct scene editing. As illustrated in Figure 5.5, simple modifications to the segmentation map, such as inserting a vehicle or removing a pedestrian, lead to corresponding visual changes in the generated image. Crucially, these edits were performed solely through the label map, with no modifications made to the clear-weather reference image. In the first output (top-right), the foggy scene includes a pedestrian and no vehicles, matching the initial semantic layout. In the second output (bottom-right), the label map was edited to include a vehicle and exclude the pedestrian, and the generated output reflects these changes with high visual fidelity and semantic consistency. This demonstrates the model’s controllability and utility in applications such as data augmentation and structured scene generation.

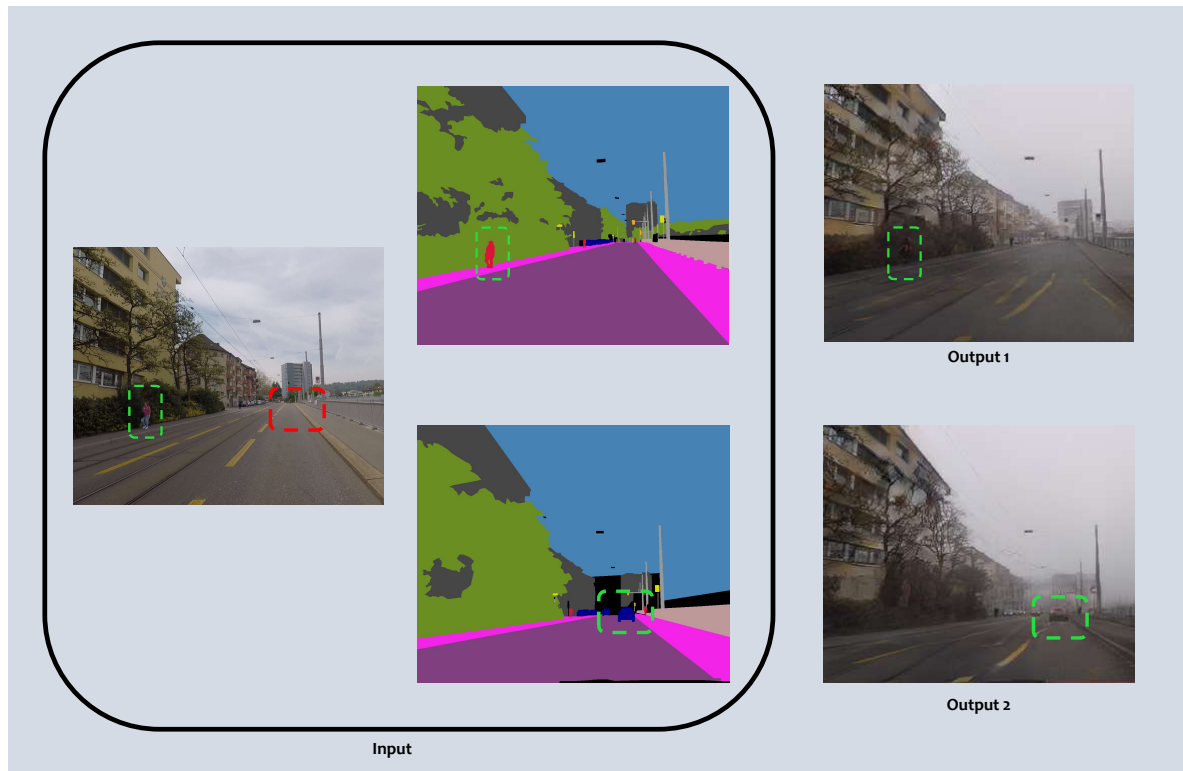


Figure 5.5: Visualization of scene-level editing using **3P-WeatherGAN**. The clear-weather reference image (left) remains unchanged, while modifications to the segmentation label map (middle) — adding a vehicle and removing a pedestrian — result in semantically consistent changes in the generated foggy outputs (right).

6 Conclusion and Future work

In this research, we proposed a novel methodology **3P-WeatherGAN** for adverse weather image synthesis using a semantic image generation framework, with a primary focus on the fog domain. Our approach introduced the use of clear-weather reference images in combination with segmentation maps to guide the generation process. Among the various strategies explored, the Three-Pyramid Dual SPADE Fusion approach—where the reference image is processed through a dedicated feature extraction pyramid—achieved the best performance in visual realism, attaining the lowest FID score across all configurations. Notably, this method outperformed other complex feature-level fusion strategies.

To evaluate the downstream utility of the generated images, we trained a semantic segmentation model using varying ratios of synthetic data produced by our best-performing GAN configuration. The results showed that the performance converged with that of the unpaired CycleGAN approach; however, our proposed methodology provided more semantic control and flexibility, which contributed positively to the task.

One promising direction for future work is to extend the proposed methodology to other weather domains such as rain, snow, or even compound weather scenarios. Each of these conditions presents unique visual challenges, and generalizing our approach across them would help validate its robustness and flexibility.

Another valuable extension would be incorporating explicit weather-related cues, such as depth maps, humidity levels, or temperature-based features, into the generation pipeline. These cues can provide additional context that may enhance the synthesis of weather-specific visual effects, leading to more accurate and controllable outputs.

Future work may also explore the introduction of novel loss functions tailored to weather conditions. In particular, a weather-aware attention loss could be used to focus the model’s learning on regions that are most affected by environmental factors, such as distant objects or reflective surfaces. This could further improve realism and domain alignment.

Finally, the growing success of generative models such as diffusion models and visual-language models (e.g., CLIP) opens new possibilities for extending our framework ???. These models enable richer control through text prompts or multi-modal conditioning, allowing for more expressive and diverse synthesis.

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Declaration

Herewith, I declare that I have developed and written the enclosed thesis entirely by myself and that I have not used sources or means except those declared.

This thesis has not been submitted to any other authority to achieve an academic grading and has not been published elsewhere.

Stuttgart, 15/05/2025

Samar Fathalah